

Writing the FYP/Master / PhD Thesis Proposal

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What is a research proposal?

- **Statement of intent**



- **Academically prepared to complete the research**
- **Audience: peers, supervisors, examiners**

What is a research proposal?

- A research proposal is your PLAN
 - It describes in detail your study
 - Decisions about your study are based on the quality of the proposal
 - Approvals to proceed by the Institutional Review Board



Importance of a research proposal?

Contract between you and your committee

1.Serves to protect the student

- Demanding additional requirements

2.Protects the committee from the student

- From delivering a degree of poor quality

What are the essential ingredients?

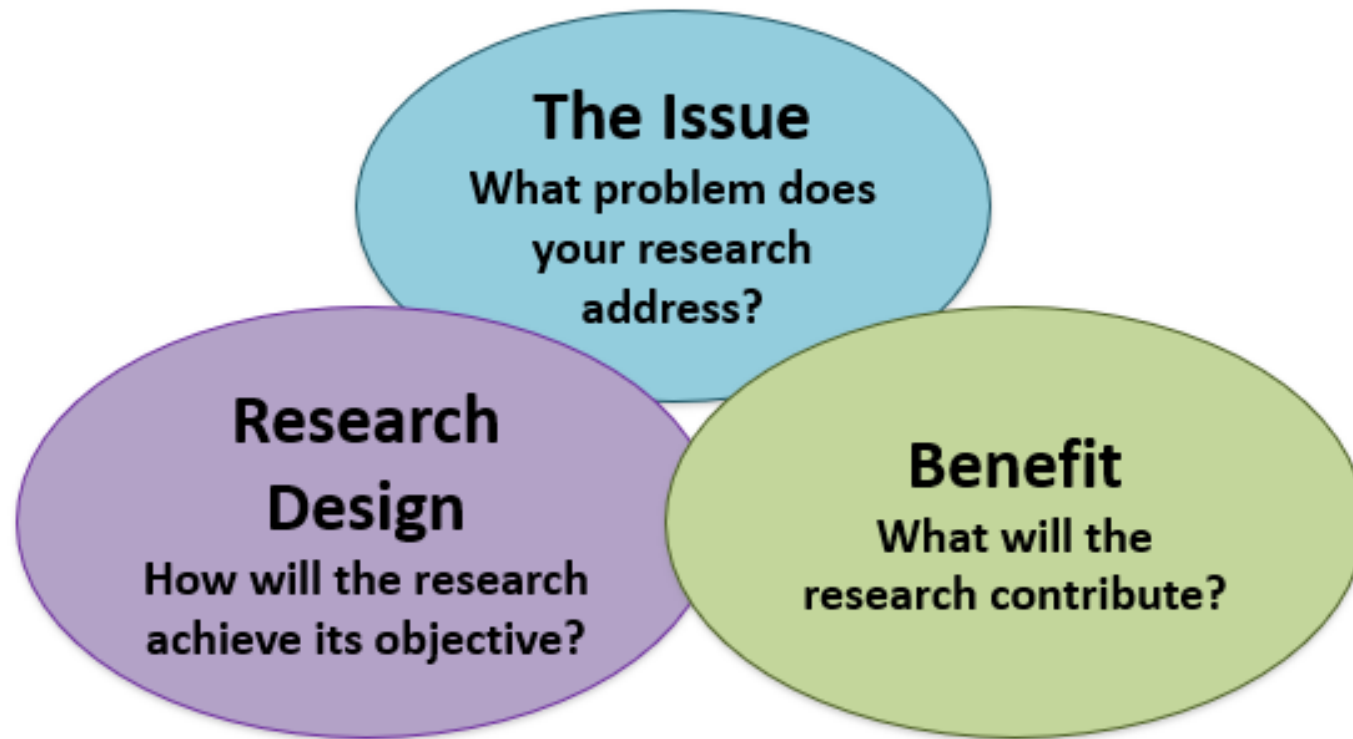


Figure adapted from MIT OCW

Research proposals make you:

OUTLINE steps in your proposed research

Provide yourself with intellectual **CONTEXT**

JUSTIFY your
research

Be **CREATIVE**

THINK through your experiments

Anticipate
potential
PROBLEMS

Anticipate a
realistic
TIMETABLE

Figure adapted from MIT OCW

Getting started

- Title
 - Outline
 - Literature Review
 - Methodology
 - Methods of data collection and analysis
 - Ethical Issues
 - Timeline
 - Resources
 - Outcomes
 - Reference list
- Title
 - Background
 - Problem statement
 - Aim and objectives
 - Rationale and context
 - Methodology
 - Plan of work
 - Resources / Support
 - Outcomes
 - Reference list

Know the requirements BEFORE you start

Title

GPS address



- Mini-abstract
- Clear
- Concise
- Subject 1st

Problem statement

- Short SO WHAT statement
- Purpose
 - Blueprint for your literature review
 - Focus your committee at the beginning
 - Keep them on track throughout your proposal



Literature review

This is NOT just a summary of literature

Show how your project:

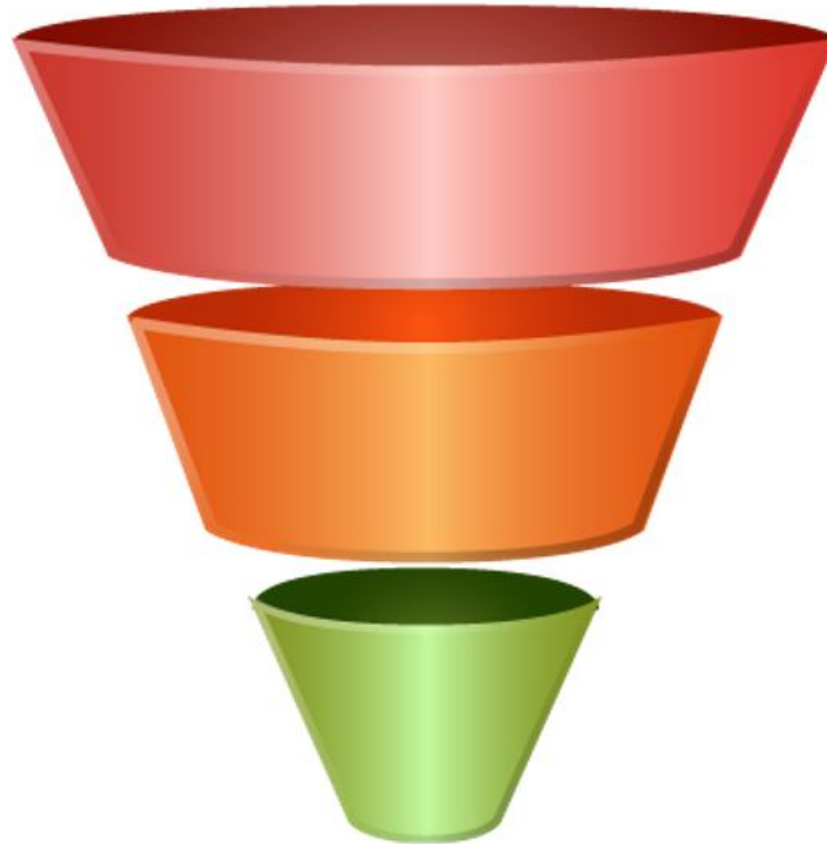
- Literature **SUPPORTS** your hypothesis
- **EXTENDS** previous work
- **AVOIDS** previous mistakes
- **IS UNIQUE** to previously followed paths

The narrative of a good literature review

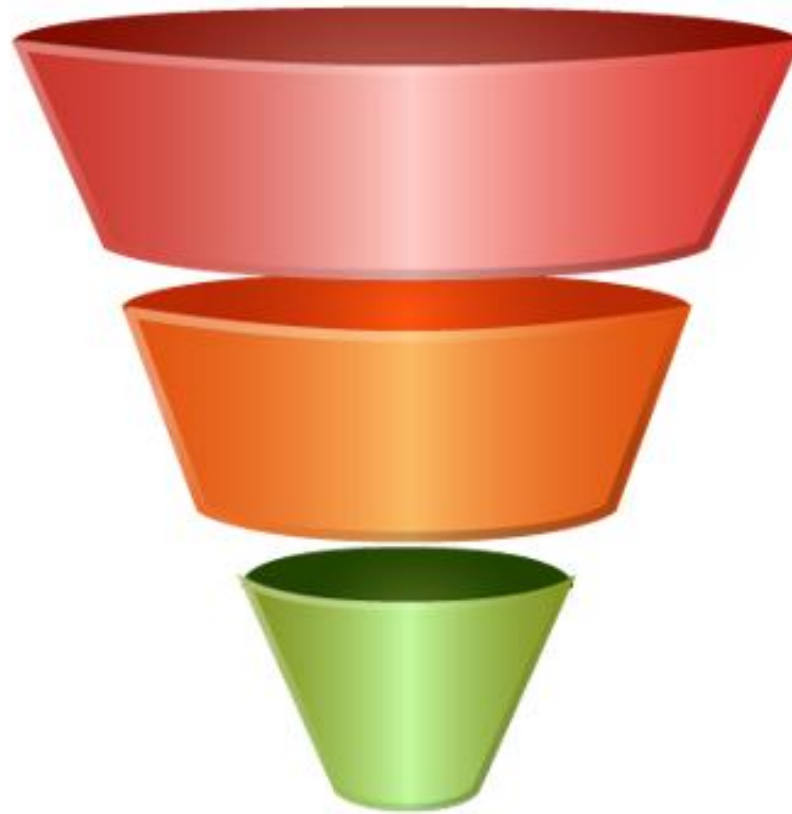
Reader
knows



Reader
doesn't
know



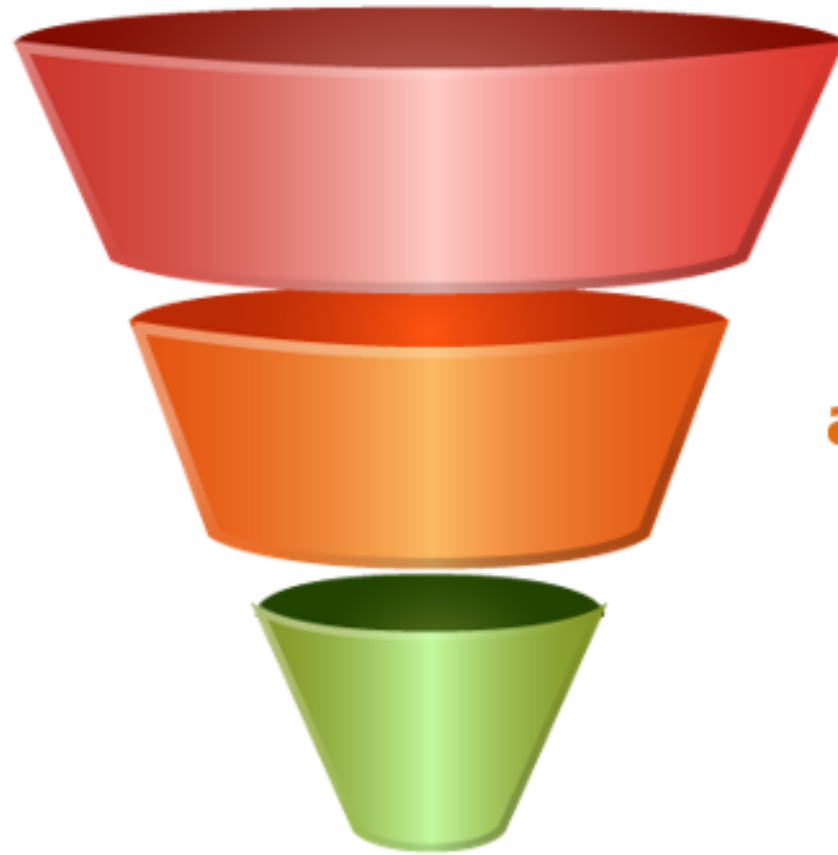
The narrative of a good literature review



**Introduce the
field:**

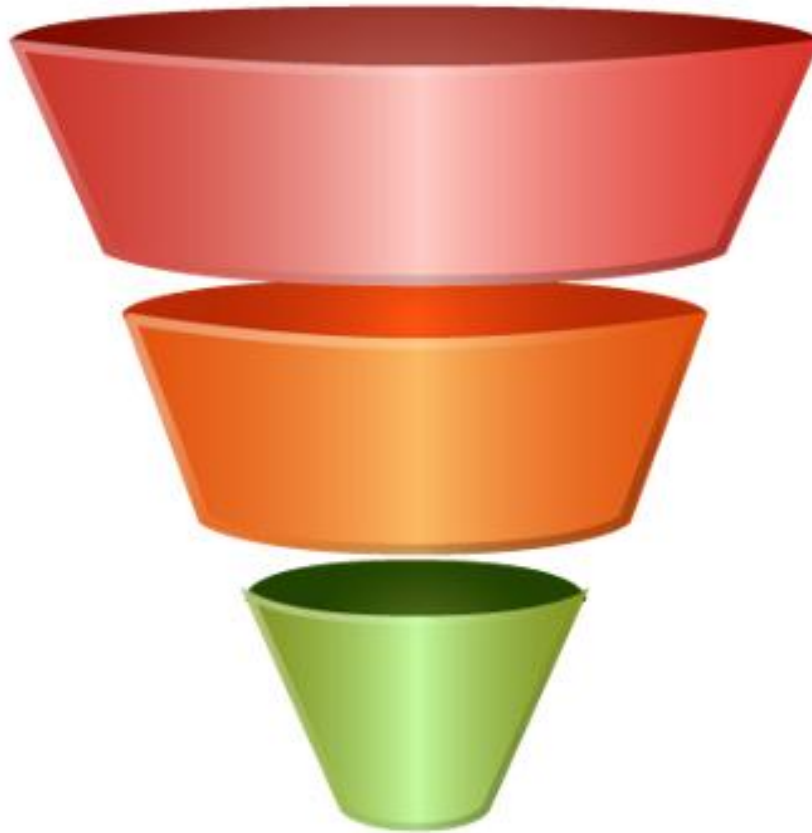
broad focus

The narrative of a good literature review



**Focus on certain
aspects in field of
interest**

The narrative of a good literature review



**End with gap
analysis**

Aims and Objectives

The purpose of this research is to.....

- **Aims**
 - short but **general** statement of intent
- **Objectives**
 - very **specific statements** that define the practical steps you will take to achieve your aim(s)

Proposed Methodology and Experiments

- Section used to **JUDGE** the validity of results and conclusions
- This section of your proposal has multiple parts
 - Instrumentation and infrastructure
 - Study groups and ethics
 - Access to samples
 - Data analysis
- Justify your method choice
- Show you understand the principles
- **Prove feasibility of your study**



Timeline



- **Helps you keep your experimental design in the correct order**
- **Avoids “dead” time**
- **When building your timeline**
 - **Consult with other students in your lab that have done similar studies**

Outcomes

- What do you expect the results to be?
- Measurable
 - E.g. you will get a degree
 - New patent / paper
- Qualitative
 - Contribute understanding to subject / new technology / application

Common pitfalls to avoid

- Forget to include table of contents
- Not connecting the proposed research to the literature review
- Not enough detail about methods
- Jargon & sweeping generalities
- Relying solely on your supervisor

Improving Your Odds

- Know what the evaluation criteria will be and **CHECK** that you meet these
- Start with an outline
- Read approved proposals
- Once “finished” ask others to proofread
- Use diagrams to illustrate models
- Make sure your mentor approves it before you submit

Evaluation of proposals

- The aims/objectives are likely to be **achievable** in the given time period
- The **rationale** for the proposed study is **reasonable**
- The scientific design is described and **adequately justified**